



L6a: Introduction to Single Index Models

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Objectives for Today

Today we replace the full covariance matrix of L5a and L5b by a model with one common factor: each asset's growth rate is a linear function of a market index plus a residual of its own.

Today's concepts

- **Decompose an asset's growth rate**: the single index model and its assumptions, what α_i , β_i , and the residual mean, and the split of variance into a market part and an asset-specific part.
- **Build the covariance a SIM implies**: rank one plus diagonal, what it saves, portfolio beta and variance, and the residual assumption that must be checked.
- **Estimate and diagnose a SIM**: least squares, classical uncertainty in growth-rate units, R^2 , and what daily data do to the assumptions.

Lectures and Examples

Lecture:

CHEME-5660-L6a-Lecture-SIM-Fall-2026.ipynb

Example 1:

CHEME-5660-L6a-Example-SVD-SIM-Estimation-Fall-2026.ipynb

Example 2:

CHEME-5660-L6a-Example-SIM-Parameter-Uncertainty-Fall-2026.ipynb

The first example fits the SIM for one firm three equivalent ways, computes its classical uncertainty next to an index fund's, fits every security in the dataset, checks the residual-correlation assumption, and saves the archive L6b loads. The second bootstraps the parameters two ways and audits the residuals.

Concept Review: Data Driven Minimum-Variance Allocation

L5b: for a portfolio $\mathcal{P}$ of risky assets with weights $\mathbf{w}$, the linearized one-period growth rate $g_p = \mathbf{w}^\top \mathbf{g}$ has reward $\mathbb{E}[g_p] = \mathbf{w}^\top \bar{\mathbf{g}}$ and risk $\text{Var}(g_p) = \mathbf{w}^\top \Sigma_g \mathbf{w}$; the long-only weights for a target $g_\star$ solve:

$$\min_{\mathbf{w}} \mathbf{w}^\top \Sigma_g \mathbf{w} \quad \text{s.t.} \quad \mathbf{w}^\top \bar{\mathbf{g}} \geq g_\star, \quad \mathbf{1}^\top \mathbf{w} = 1, \quad w_i \geq 0.$$

Sweeping the floor traces the GMV portfolio and the efficient branch.

Every entry is an estimate: $\bar{\mathbf{g}}$ and Σ_g , one covariance per pair of assets, and L5b's advanced material showed how noisy those pairwise entries are and how the optimizer amplifies the noise. **Today's question:** can a model with far fewer parameters supply the same inputs?

Catch-up: [Archived data-driven minimum-variance example](#)

Factor Models

A portfolio of $|\mathcal{P}|$ assets needs $|\mathcal{P}|$ means and $|\mathcal{P}|(|\mathcal{P}| + 1)/2$ covariance entries; the second grows with the square of the first. Yet securities move together when market-wide news arrives. A **factor model** makes the shared movement explicit. Let $g_{i,t}$ be asset i 's growth rate on day t (inverse years), $g_{M,t}$ the growth rate of a market index M (SPY here; the letter names the index), $f_{k,t}$ further factors, α_i an intercept (inverse years), β_i the sensitivity to the market (dimensionless), b_{ik} the sensitivity to factor k , and $\varepsilon_{i,t}$ the residual (inverse years):

$$g_{i,t} = \underbrace{\alpha_i}_{\text{intercept}} + \underbrace{\beta_i g_{M,t}}_{\text{market}} + \overbrace{\sum_{k=1}^K b_{ik} f_{k,t}}^{\text{other factors}} + \varepsilon_{i,t}.$$

Co-movement between assets is carried by the factors; the residuals are what is left over.

Two Named Factor Models, and the One We Keep

- **CAPM (bridge from L5b)**: an equilibrium statement about expected excess *simple* returns, no intercept. In annualized growth rates with a risk-free growth rate g_f the CAPM-style relation is $g_{i,t} - g_f \approx \beta_i(g_{M,t} - g_f) + \varepsilon_{i,t}$: an approximation, and a time-series regression keeps an intercept whose CAPM value would be zero. We borrow the one-factor structure, not the test.
- **Fama–French**: excess market simple return $R_M - R_f$ plus a size factor (SMB) and a value factor (HML); targets common variation the market factor omits, at the cost of more parameters.
- **Single index model**: the market factor alone. Today's model.

Single Index Models (SIMs)

Sharpe (1963). For asset $i \in \mathcal{P}$ on days $t = 1, \dots, N$, with intercept α_i (inverse years), market exposure β_i (dimensionless), and residual $\varepsilon_{i,t}$ (inverse years):

$$g_{i,t} = \alpha_i + \beta_i g_{M,t} + \varepsilon_{i,t}$$

α_i : projection at $g_M = 0$; $\varepsilon_{i,t}$: the **idiosyncratic** remainder. A conditional-mean reading additionally needs $\mathbb{E}[\varepsilon_{i,t} \mid g_{M,t}] = 0$. Population assumptions:

$$\mathbb{E}[\varepsilon_{i,t}] = 0, \quad \text{Cov}(\varepsilon_{i,t}, g_{M,t}) = 0, \quad \text{Cov}(\varepsilon_{i,t}, \varepsilon_{j,t}) = 0 \quad (i \neq j).$$

- If $\text{Var}(g_M) > 0$, the first two are projection identities for α_i and β_i .
- The third is an **extra restriction**: the market is the only source of co-movement. Separate fits do not make it true; L6b's covariance rests on it; Example 1 checks it.
- Return form $r = \Delta t g$: same β_i ; $\alpha_{r,i}$ and $\varepsilon_{r,i,t}$ are each Δt times their growth-rate forms.

What Beta Means

Covariance of both sides with g_M (the intercept is a constant, $\text{Cov}(\varepsilon_i, g_M) = 0$). With $\sigma_{g,M}$, $\sigma_{g,i}$ the market's and asset's growth-rate standard deviations (L3a's σ_g , not volatilities; both positive) and ρ_{iM} their correlation:

$$\beta_i = \frac{\text{Cov}(g_i, g_M)}{\text{Var}(g_M)} = \rho_{iM} \frac{\sigma_{g,i}}{\sigma_{g,M}} \quad \alpha_i = \mathbb{E}[g_i] - \beta_i \mathbb{E}[g_M].$$

- Beta is the slope of the projection: a one-unit market move changes linearly projected asset growth by β_i units; the intercept and the residual are there on every day.
- **Dial:** $\beta_i = 1$ one for one; > 1 amplifies; $0 < \beta_i < 1$ dampens; < 0 moves against the market. In our dataset betas run from near zero to a little above two, median close to one.

Risk Splits in Two

Variance of both sides, using only zero covariance between ε_i and g_M (independence not needed), with $\sigma_{g,\varepsilon,i}^2 = \text{Var}(\varepsilon_i)$:

$$\underbrace{\text{Var}(g_i)}_{\text{total}} = \underbrace{\beta_i^2 \sigma_{g,M}^2}_{\text{systematic}} + \underbrace{\sigma_{g,\varepsilon,i}^2}_{\text{idiosyncratic}}$$

- Systematic risk is inherited from the market and cannot be diversified away by adding assets that share it; idiosyncratic risk is the asset's own.
- The two need not be comparable: a stock can have $\beta > 1$ and be dominated by its residual; an index fund $\beta \approx 1$ and almost none. Thus beta is exposure, not total risk: here $\beta_i > 1$ implies greater total variance, but $0 < \beta_i < 1$ need not imply less because residual variance may dominate.
- The fraction the market explains is $\beta_i^2 \sigma_{g,M}^2 / \text{Var}(g_i) = \rho_{iM}^2$: what the fitted R^2 estimates (for a non-constant asset growth rate).

The Covariance Implied by a SIM

Stack the assets: growth rates $\mathbf{g}$, intercepts α , betas β , residuals ε ; let $\bar{g}_M = \mathbb{E}[g_M]$, $\sigma_{g,M}^2 = \text{Var}(g_M)$, and $\mathbf{D}_g = \text{diag}(\sigma_{g,\varepsilon,1}^2, \sigma_{g,\varepsilon,2}^2, \dots)$. The vector model $\mathbf{g} = \alpha + \beta g_M + \varepsilon$ gives, under the three assumptions:

$$\bar{\mathbf{g}} = \alpha + \beta \bar{g}_M, \quad \Sigma_g = \sigma_{g,M}^2 \beta \beta^\top + \mathbf{D}_g$$

so $\text{Cov}(g_i, g_j) = \beta_i \beta_j \sigma_{g,M}^2$ off the diagonal and $\text{Var}(g_i) = \beta_i^2 \sigma_{g,M}^2 + \sigma_{g,\varepsilon,i}^2$ on it.

- **Rank at most one plus diagonal:** every off-diagonal covariance is two betas times one number.
- The compression: $2|\mathcal{P}| + 1$ numbers (one beta and one residual variance per asset, plus the market variance) instead of $|\mathcal{P}|(|\mathcal{P}| + 1)/2$ covariances, plus $|\mathcal{P}| + 1$ for $\bar{\mathbf{g}}$. For the S&P 500, on the order of a thousand parameters against more than a hundred thousand.

What Is Assumed Away

The zeros off the diagonal of $\mathbf{D}_g$ are the third assumption. If two assets' residuals are correlated, their true covariance is $\beta_i \beta_j \sigma_{g,M}^2$ plus a residual covariance the model sets to zero.

Fitting each regression separately makes every residual orthogonal to the market column, and nothing else. Example 1 measures the residual correlations across our dataset:

- most pairs close to zero (median absolute correlation below 0.1),
- related firms a few tenths (banks in the portfolio example reach about 0.7),
- near-duplicate securities almost perfectly correlated.

Those are the covariance errors L6b's model inherits: the number to quote when asked how good the SIM covariance is.

Portfolio Risk Under a SIM

For fixed weights $\mathbf{w}$, the linearized growth rate $g_p = \mathbf{w}^\top \mathbf{g}$ has portfolio beta $\beta_p = \mathbf{w}^\top \beta$ and residual $\varepsilon_p = \mathbf{w}^\top \varepsilon$:

$$g_p = \mathbf{w}^\top \alpha + \beta_p g_M + \varepsilon_p, \quad \text{Var}(g_p) = \underbrace{\beta_p^2 \sigma_{g,M}^2}_{\text{market risk}} + \underbrace{\mathbf{w}^\top \mathbf{D}_g \mathbf{w}}_{\text{residual risk}}$$

- Market risk depends on the weights only through β_p ; adding assets with the same exposure does not reduce it.
- Residual risk diversifies: equal weights $1/|\mathcal{P}|$ over $|\mathcal{P}|$ assets with residual variances below K give $\mathbf{w}^\top \mathbf{D}_g \mathbf{w} = \frac{1}{|\mathcal{P}|^2} \sum_i \sigma_{g,\varepsilon,i}^2 \leq K/|\mathcal{P}|$.
- Two conditions carry it: spread-out fixed weights, and **uncorrelated residuals** (with a general residual covariance Ω_ε the term is $\mathbf{w}^\top \Omega_\varepsilon \mathbf{w}$ and need not vanish).

Estimation of SIM Parameters

One asset at a time, N aligned daily observations (the L5a data-matrix columns). Response $\mathbf{y} = (g_{i,1}, \dots, g_{i,N})^\top$; design matrix $\hat{\mathbf{X}} \in \mathbb{R}^{N \times 2}$ with a column of ones and the market column $g_{M,1}, \dots, g_{M,N}$; parameters $\theta_i = (\alpha_i, \beta_i)^\top$; residuals ε : $\mathbf{y} = \hat{\mathbf{X}}\theta_i + \varepsilon$.

$$\hat{\theta}_i = \arg \min_{\theta} \|\mathbf{y} - \hat{\mathbf{X}}\theta\|_2^2 \implies (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})\hat{\theta}_i = \hat{\mathbf{X}}^\top \mathbf{y} \implies \hat{\theta}_i = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \mathbf{y}$$

unique when $\hat{\mathbf{X}}$ has full column rank. Fitted values $\hat{\mathbf{y}} = \hat{\mathbf{X}}\hat{\theta}_i$, residuals $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}}$; $\hat{\beta}_i$ is the sample covariance over the sample market variance, $\hat{\alpha}_i = g'_i - \hat{\beta}_i g'_M$.

In code: never form the inverse; a QR solve ($\mathbf{X} \backslash \mathbf{y}$) or the SVD gives the same estimate stably (Example 1 does it three ways). Ridge versions: optional advanced material.

Example 1: CHEME-5660-L6a-Example-SVD-SIM-Estimation-Fall-2026.ipynb

Uncertainty

Classical assumptions: design held fixed, $\mathbb{E}[\varepsilon \mid \hat{\mathbf{X}}] = \mathbf{0}$, errors independent across days, common variance, Gaussian, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma_{g,\varepsilon}^2 \mathbf{I})$. Then $\hat{\theta}_i = \theta_i + (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \varepsilon$ is unbiased and Gaussian with covariance $\sigma_{g,\varepsilon}^2 (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}$. Observations are already growth rates: **no Δt** .

$$s_{g,\varepsilon}^2 = \frac{\|\mathbf{r}\|_2^2}{N-2}, \quad \widehat{\text{Cov}}(\hat{\theta}_i) = s_{g,\varepsilon}^2 (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1}, \quad \text{SE}(\hat{\theta}_j) = \sqrt{[\widehat{\text{Cov}}(\hat{\theta}_i)]_{jj}}$$

95% interval: $\hat{\theta}_j \pm t_{0.975, N-2} \text{SE}(\hat{\theta}_j)$ (1.96 for eleven years of daily data). Beta's interval is usually narrow; alpha's usually wide and often covering zero. Residual diffusion volatility: $\sqrt{\Delta t} s_{g,\varepsilon}$.

Fit, and What Daily Data Do to the Assumptions

$$R^2 = 1 - \frac{\|\mathbf{r}\|_2^2}{\sum_t (g_{i,t} - g'_i)^2} = \hat{\rho}_{iM}^2 \quad (\text{least squares, intercept, non-constant response})$$

An index fund can sit near 0.85 against SPY, a stock with a larger beta near 0.25: both consistent with the model.

- Daily growth rates: heavy tails, clustered absolute values, and (from volume-weighted average prices) positive lag-one autocorrelation: the classical beta interval is too narrow for these data; a Newey–West standard error is 40 to 100% larger.
- **Bootstrap:** $\mathbf{y}^{(b)} = \hat{\mathbf{X}}\hat{\theta}_i + \varepsilon^{(b)}$, refit, read the spread; parametric (Gaussian) or empirical-residual (keeps the tails); both draw days independently. When dependence matters: HAC errors or a block bootstrap. Example 2 does all of it.

Example 2: CHEME-5660-L6a-Example-SIM-Parameter-Uncertainty-Fall-2026.ipynb

What R-squared, Alpha, and Beta Do Not Prove

- R^2 is the sample fraction of variation the fitted market term explains: not the probability the model is true, not forecast accuracy, not investment quality. Low R^2 with a precise beta, high R^2 with unstable parameters: both happen.
- A positive $\hat{\alpha}_i$ is an intercept conditional on the market proxy, the window, the price series (volume-weighted average prices, not total returns), and the specification; its interval usually covers zero.
- Beta is exposure to the chosen index at the chosen sampling interval, not total risk.
- Before the SIM covariance goes into L6b's optimizer: residual cross-correlations, stability across windows, sensitivity to the proxy, and one unit convention throughout (growth rate, log return, covariance rate differ by powers of Δt).

Optional Advanced Material

Advanced: `advanced/sim/`

`CHEME-5660-L6a-Advanced-SIM-Theory-Fall-2026.ipynb`

Optional, not a prerequisite for L6b. Derives the ridge (regularized) estimator and its covariance, distinguishes the residual and parametric bootstraps, keeps the growth-rate, log-return, and volatility conventions straight, propagates parameter uncertainty into portfolio risk and weights, and writes the maximum-Sharpe problem as a second-order cone program.

Notebook: [lectures/week-6/L6a/advanced](#), also linked from the lecture notebook.

Summary

Today: the single index model, the covariance and portfolio risk it implies, and least-squares estimation with classical uncertainty.

Key takeaways

- **One factor splits risk in two:** beta squared times the market variance plus a residual variance; beta measures exposure, R^2 what the market explains, and they can disagree.
- **The SIM covariance is rank one plus diagonal, on an assumption:** one beta and one residual variance per asset; residual risk diversifies only if residuals are uncorrelated, which fits do not guarantee.
- **Least squares estimates, daily data qualify:** growth-rate units, no Δt ; heavy tails, lag-one dependence, and clustering make bootstrap or robust intervals the complement.

Next: the SIM mean and covariance inside L5b's allocation problems.

Today: one index, one beta per asset, and the covariance that falls out of a single line.